pH Meter



Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **Medium** 6 labs

Covers these lab items: pH meter; Ph Paper Strips/Small pH Meter

What it is used for

A pH meter measures how acidic or alkaline a liquid sample is by detecting the response of a pH electrode. Proper calibration, rinsing, and storage of the electrode help produce reliable measurements.

Main parts

- pH meter display and control buttons
- pH electrode or combined probe
- Temperature sensor or automatic temperature compensation feature if fitted
- Electrode storage solution and protective cap
- Calibration buffer solutions
- Sample beakers, rinse bottle, and lint-free tissue

Before you start

1. Read the instrument instructions and confirm the expected pH range of the sample.
2. Inspect the meter, electrode cable, probe body, and connector for damage.
3. Check the electrode condition and ensure its sensing bulb or junction is not dry or cracked.
4. Prepare fresh, suitable calibration buffers that bracket the expected sample pH.
5. Switch on the meter and allow it to stabilize as specified by the manufacturer.
6. Wear suitable gloves and eye protection when handling samples or buffer chemicals.

How to use

1. Rinse the electrode with distilled or deionized water and gently blot it without rubbing the sensing bulb.
2. Calibrate the meter using the required buffer solutions and follow the displayed prompts.
3. Rinse the electrode between buffer solutions to prevent cross-contamination.
4. Pour a representative sample into a clean beaker without contaminating the main sample container.
5. Immerse the electrode and temperature probe to the recommended depth without touching the beaker bottom.
6. Gently stir if required and wait for the reading to stabilize.
7. Record the pH value, sample identification, temperature, and any required measurement conditions.
8. Rinse the electrode after measurement and repeat the test if the result is unstable or outside expected limits.

9. Store the electrode in the recommended storage solution and replace its protective cap.
10. Switch off the meter and clean the work area after testing.

⚠ Safety

- Never handle electrodes by pulling their cables or strike the glass sensing bulb against hard surfaces.
- Wear suitable protection when testing corrosive, toxic, or biologically hazardous samples.
- Do not store the electrode in distilled water unless the manufacturer specifically permits it.
- Keep electrical connectors dry and disconnect the meter before cleaning non-waterproof components.
- Do not return tested samples to their original containers.
- Dispose of buffer solutions and samples according to laboratory procedures.

Care & storage

- Rinse the electrode after every sample and remove deposits using approved cleaning solution.
- Store the electrode in the specified solution and keep its sensing element hydrated as instructed.
- Calibrate at the required frequency and whenever results or conditions indicate drift.
- Inspect electrode response, cable condition, and meter battery or power supply regularly.

Common mistakes to avoid

- Skipping calibration can produce inaccurate pH readings.
- Wiping the sensing bulb harshly can create static effects or damage the electrode.
- Storing the electrode dry or in an unsuitable liquid can shorten its service life.
- Using contaminated or expired buffers can make calibration unreliable.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.